

Notes: Cai, He, Jiang & Xiong (RFS, 2021)
The Whack-a-Mole Game: Tobin Taxes and Trading Frenzy

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1 Big picture

- **Question.** What happens when regulators raise a Tobin tax in one market while close speculative substitutes remain untaxed?
- **Policy shock.** On May 30, 2007, the Chinese government tripled the stamp tax on stock trading from 0.2% to 0.6% to cool a rapidly rising stock market.
- **Institutional feature.** The tax applied to stocks but not to exchange-traded stock warrants. Warrants were therefore a nearby untaxed speculative venue.
- **Main result.** The tax increase modestly cooled stock-market turnover, but trading frenzy migrated into the warrant market and intensified the warrant bubble.
- **Investor mechanism.** Account-level data show both inflows of new warrant investors and large increases in trading intensity by existing warrant investors.
- **Warrant-market effect.** Deep out-of-the-money put warrants, already known to have little fundamental value, experienced sharp increases in prices, turnover, volume, and volatility.
- **Call-warrant effect.** Call warrants also experienced increases in turnover and volatility, with stronger responses among warrants that were more speculative before the event.
- **Regulatory lesson.** A narrowly targeted transaction tax can create a whack-a-mole problem: speculation is suppressed in the taxed market but reappears in related unregulated markets.
- **Economic takeaway.** The effectiveness of a Tobin tax depends on the full opportunity set of investors. If substitute trading venues exist, raising costs in one market can shift and amplify speculative activity elsewhere.

2 Incremental contribution of the paper

- **From direct-market effects to cross-market migration.** Earlier Tobin-tax studies mostly ask whether transaction taxes reduce volatility or turnover in the taxed market. This paper studies whether trading frenzy relocates to an untaxed substitute market.
- **Clean policy event.** The May 30, 2007, stamp-tax hike was large, sudden, and specific to stock trading. Warrants were not subject to the tax, which creates a natural contrast between taxed and untaxed speculative venues.
- **Use of account-level evidence.** The paper goes beyond aggregate prices and volumes by using investor account data to identify new warrant investors, existing warrant investors, and investor groups by trading intensity.
- **Mechanism evidence.** It shows that the warrant-market response is not merely mechanical substitution by new entrants. Existing active warrant investors also increased trading sharply, consistent with speculative feedback and greater-fool dynamics.
- **Policy contribution.** The paper provides concrete evidence for a regulatory whack-a-mole channel: market participants may sidestep regulation by moving into instruments outside the regulatory perimeter.

3 Data and measurement

3.1 Markets and assets

- The study focuses on China's A-share stock market and the listed warrant market around May 30, 2007.
- The warrant sample contains 14 active warrants: five put warrants and nine call warrants.

- All warrants had long maturities of one to two years and were traded on either the Shanghai Stock Exchange or the Shenzhen Stock Exchange.
- Put warrants were deep out-of-the-money and had virtually no fundamental value, which makes them a clean setting for measuring speculative trading.

Interpretation: The key comparison is between a large stock-market tax increase and an untaxed derivatives market that offered a nearby outlet for speculative demand.

3.2 Event window

- The main event date is May 30, 2007, when the stamp tax increase took effect.
- The baseline event window is 20 trading days before and 20 trading days after May 30, from April 25 to June 26, 2007.
- The paper also uses placebo dates when the stock index fell more than 5% to distinguish tax-induced migration from generic stock-market crashes.

Interpretation: The 20-day window keeps the analysis close to the shock while still allowing enough observations to compare market behavior before and after the tax change.

3.3 Market-level variables

- **Return:** daily holding return adjusted for dividends and splits.
- **Turnover:** shares traded divided by floating shares.
- **Volume:** daily trading volume, measured in billion yuan for stocks and yuan for warrants.
- **log(Volume):** log of one plus daily trading volume.
- **Volatility:** daily high-low price range scaled by average price.
- **Adjusted price:** warrant price net of Black-Scholes value, used to measure speculative overpricing.

Interpretation: The variables are designed to distinguish price effects, trading intensity, trading value, and volatility. For warrants, adjusted price is especially important because raw warrant prices include both fundamentals and bubble components.

3.4 Investor-account variables

- The paper uses brokerage account data from the Shenzhen Stock Exchange.
- **Number of trades** counts how frequently investors trade a warrant.
- **Number of purchases** and **Number of sales** separate buying and selling pressure.
- **Total value of purchases** and **Total value of sales** measure yuan trading intensity.
- **Number of investors** counts the investors who trade a warrant on a given day or event window.
- Investors are sorted into activeness groups based on pre-event trading intensity.

Interpretation: Account data allow the authors to separate entry by new investors from intensified speculation by existing investors. This is the main mechanism evidence in the paper.

4 Models and empirical specifications

4.1 Event-study regression

For daily market variables, the baseline specification is:

$$Y_{j,t} = \alpha + \beta Post_t + \varepsilon_{j,t},$$

where $Y_{j,t}$ is a stock- or warrant-level variable and $Post_t$ equals one after May 30, 2007.

Interpretation: The coefficient β measures the average change in the outcome after the tax increase relative to the pre-event period. For stocks, a negative turnover coefficient indicates cooling of the taxed market. For warrants, positive coefficients indicate migration of speculative activity.

4.2 Call-warrant heterogeneity specification

For call warrants, the paper tests whether more speculative warrants before the event respond more strongly:

$$DepVar_{j,t} = \alpha + \beta_1 Post_t + \beta_2 PreEventTurnover_j + \beta_3 (Post_t \times PreEventTurnover_j) + \delta_m + \varepsilon_{j,t}.$$

Interpretation: A positive β_3 means that warrants with greater pre-event speculative trading experience larger post-tax increases. This supports the idea that the tax shock redirected speculative demand toward the most attractive speculative substitutes.

4.3 Investor-entry and investor-intensity regressions

The investor-level event analysis compares the number and intensity of warrant investors before and after May 30:

$$InvestorOutcome_{g,t} = \alpha_g + \beta Post_t + \varepsilon_{g,t},$$

where g can index investor groups by pre-event activeness.

Interpretation: The distinction between new investors and existing active investors is central. If only new investors enter, the mechanism is substitution from stocks to warrants. If existing warrant traders also increase intensity, the shock amplified speculative interaction inside the warrant market.

4.4 Overnight-return mechanism

To distinguish overnight price pressure from daytime trading pressure, the paper studies:

$$OvernightReturn = \frac{Open_t - Close_{t-1}}{Close_{t-1}},$$

and uses implied deltas to translate warrant returns into stock-equivalent returns.

Interpretation: Large positive overnight returns in put warrants on May 30 indicate demand pressure at the open. Because these put warrants had very low fundamental values, these overnight movements are hard to explain by fundamentals and point to speculative inflows.

5 Identification and empirical design

5.1 Policy shock

The core identifying event is the sudden tripling of the stamp tax on stock trading on May 30, 2007. The shock is useful because it was large, salient, and applied to stocks but not to warrants.

Interpretation: The design does not require that the stock market itself be unaffected by other events. The key contrast is that investors faced a sharp increase in the cost of stock trading while a nearby speculative derivative market remained untaxed.

5.2 Taxed market versus untaxed substitute market

The empirical comparison has two sides:

- In the **taxed stock market**, turnover and index behavior show whether the tax cooled the target market.
- In the **untaxed warrant market**, prices, turnover, and investor entry show whether speculative activity migrated.

Interpretation: This is the whack-a-mole logic. The target market may appear to cool, but the policy can still fail if speculative trading simply moves to a substitute venue.

5.3 Investor-account validation

Aggregate warrant prices could rise for many reasons. The account data validate the mechanism by showing who traded more. The paper separates:

- investors who entered warrants for the first time after the tax increase;
- existing warrant traders who increased their trading intensity;
- highly active warrant traders, who are most likely to amplify speculative dynamics.

Interpretation: This evidence is important because the paper's interpretation is behavioral and regulatory. The tax did not just change one price; it changed investor behavior across markets.

5.4 Placebo-event design

The paper compares the May 30 event with other days when the stock market fell by more than 5%. These placebo crashes help distinguish the tax effect from generic flight into warrants after stock-market declines.

Interpretation: If warrant trading always surged after big stock-market drops, May 30 would be less special. The placebo evidence shows that the combination of a large stock-market fall and a tax increase produced a distinctive warrant-market response.

5.5 Limitations

- The event is not randomized and is a single policy episode.
- The warrant market was unusual because warrants were heavily retail traded and short-sale constraints limited arbitrage.
- The policy effect may not generalize to all Tobin taxes or all derivative markets.

- The event window is short, so the paper identifies immediate migration and amplification rather than long-run welfare effects.

Interpretation: The strongest claim is not that Tobin taxes always worsen bubbles. The strongest claim is that narrowly targeted transaction taxes can create cross-market spillovers when speculative substitutes exist.

6 Tables and figures

6.1 Figure 1: Market turnover, index level, and stamp tax rate over time; Figure 2: Daily market turnover and index level around May 30, 2007

Read:

- Figure 1 places the 2007 tax increase in historical context: China's stock market had experienced large booms and busts, and the stamp tax was repeatedly adjusted as a regulatory tool.
- Figure 2 zooms into the event window. The stock index drops around May 30 and stock-market turnover falls after the tax hike.
- The figures show that the policy achieved a modest cooling of the target market.

Economic message: The immediate stock-market reaction is consistent with regulators partially reducing activity in the taxed market, but it does not reveal where speculative demand went next.

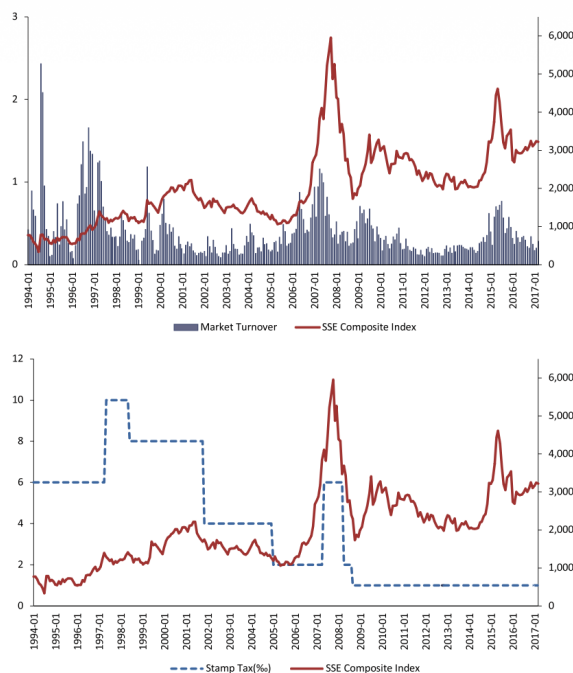


Figure 1
Market turnover, index level, and stamp tax rate over time
The upper panel plots the month-end level of the Shanghai Stock Exchange Composite index (SSE Composite index, right y-axis) and the monthly turnover rate (left y-axis). The lower panel plots the month-end level of the SSE Composite index (right y-axis) and the stamp tax rate (as 1%, left y-axis). The sample period is from January 1994 to December 2016.

(a) Figure 1: Market turnover, index level, and stamp tax rate over time.

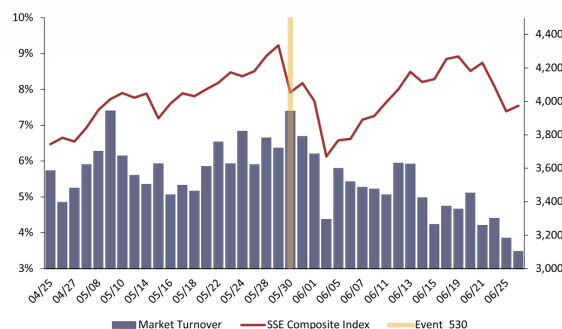


Figure 2
Daily market turnover and index level around May 30, 2007
The figure plots the daily market turnover (left y-axis) and the level of the Shanghai Stock Exchange Composite index (SSE Composite index, right y-axis) from April 25, 2007, to June 26, 2007.

(b) Figure 2: Daily stock-market turnover and SSE index around the tax event.

6.2 Table 1: List of all May 30, 2007, warrants; Table 2: Stock price and trading around May 30, 2007

Read:

- Table 1 lists the 14 active warrants in the sample, including put and call warrants on the Shanghai and Shenzhen exchanges.
- Table 2 shows that stock returns and turnover fell after the tax event, while volatility rose.

- The post-event turnover coefficient is negative, consistent with the stamp tax lowering stock-market trading intensity.

Economic message: These two tables define the asset universe and establish the baseline: the tax affected stocks, while warrants were nearby untaxed instruments available for substitution.

Table 1
List of all May 30, 2007, warrants

Warrant code	Warrant name	Type	Exchange	Trading period		Exercise period	
				Begin	End	Begin	End
030002	Wuliang YGC1	Call	SZSE	Apr. 3, 2006	Mar. 26, 2008	Mar. 27, 2008	Apr. 2, 2008
031001	Qiaocheng HQC1	Call	SZSE	Nov. 24, 2006	Nov. 16, 2007	Nov. 19, 2007	Nov. 23, 2007
031002	Gangfan GFC1	Call	SZSE	Dec. 12, 2006	Dec. 4, 2008	Nov. 18, 2008	Dec. 11, 2008
038003	Hualing JTP1	Put	SZSE	Mar. 2, 2006	Feb. 22, 2008	Feb. 27, 2008	Feb. 29, 2008
038004	Wuliang YGP1	Put	SZSE	Apr. 3, 2006	Mar. 26, 2008	Mar. 27, 2008	Apr. 2, 2008
038006	Zhongji ZYP1	Put	SZSE	May 25, 2006	Nov. 16, 2007	Nov. 19, 2007	Nov. 23, 2007
038008	Jiafei JTP1	Put	SZSE	Jun. 30, 2006	Jun. 22, 2007	Jun. 25, 2007	Jun. 29, 2007
580008	Guodian JTB1	Call	SSE	Sep. 5, 2006	Aug. 28, 2007	Aug. 29, 2007	Sept. 4, 2007
580009	Yili CWB1	Call	SSE	Nov. 15, 2006	Nov. 7, 2007	Nov. 8, 2007	Nov. 14, 2007
580010	Magang CWB1	Call	SSE	Nov. 29, 2006	Nov. 14, 2008	Nov. 15, 2007	Nov. 28, 2008
580011	Zhonghua CWB1	Call	SSE	Dec. 18, 2006	Dec. 10, 2007	Dec. 11, 2007	Dec. 17, 2007
580012	Yunhua CWB1	Call	SSE	Mar. 8, 2007	Feb. 20, 2009	Feb. 23, 2009	Mar. 6, 2009
580013	Wugang CWB1	Call	SSE	Apr. 17, 2007	Apr. 9, 2009	Apr. 10, 2009	Apr. 16, 2009
580997	Zhaoshang CMP1	Put	SSE	Mar. 2, 2006	Aug. 24, 2007	Aug. 27, 2007	Aug. 31, 2007

This table shows the summary information for the 14 warrants in our sample. The sample includes all warrants traded over the 20 trading days before and after May 30, 2007, that is, from April 25, 2007, to June 26, 2007. For each warrant, the table lists its code, name, type (call or put), exchange (SSE, Shanghai Stock Exchange, or SZSE, Shenzhen Stock Exchange), trading period, and exercise period. Warrants are sorted at the beginning of their trading period.

(a) Table 1: List of active warrants.

Table 2
Stock price and trading around May 30, 2007

A. Summary statistics								
	Mean	SD	P1	P25	P50	P75	P99	Obs.
Before May 30, 2007								
Return	1.36%	5.93%	-6.49%	-1.38%	0.98%	3.57%	10.00%	25,712
Turnover	7.46%	4.19%	0.61%	4.73%	6.82%	9.44%	20.80%	25,712
Volume (bn yuan)	0.234	0.346	0.011	0.072	0.138	0.261	1.726	25,712
log(Volume)	18.74	1.06	16.2	18.1	18.75	19.38	21.27	25,712
Volatility	5.78%	2.67%	0.00%	3.93%	5.32%	7.22%	13.60%	25,712
After May 30, 2007								
Return	-0.98%	6.25%	-10.00%	-5.00%	-0.54%	3.02%	10.00%	26,606
Turnover	6.55%	3.91%	0.20%	4.05%	6.05%	8.35%	19.40%	26,606
Volume (bn yuan)	0.208	0.328	0.002	0.057	0.112	0.226	1.635	26,606
log(Volume)	18.53	1.179	14.61	17.87	18.54	19.24	21.22	26,606
Volatility	7.78%	3.60%	0.00%	5.08%	7.42%	10.20%	16.90%	26,606
B. Regression results								
Dep. variable:	(1)	(2)	(3)	(4)				
	Return	Turnover	log(Volume)	Volatility				
Post-event	-0.0233 (-2.37)	-0.0091 (-3.03)	-0.208 (-3.18)	0.0200 (4.13)				
Observations	52,318	52,318	52,318	52,318				
Adjusted R^2	.035	.012	.009	.090				

This table reports summary statistics (panel A) and the results of regression analyses (panel B) of stocks' daily return, turnover, volume, and price volatility before and after the event date of May 30, 2007. *Post-event* equals one for days on or after May 30, 2007, and otherwise zero. *Return* refers to daily return adjusted for dividends and splits. *Volume* is daily trading volume in billion yuan. *log(Volume)* is the log of one plus daily trading volume (in yuan). *Turnover* equals the number of shares traded on each day divided by the number of floating shares. *Volatility* is measured on a daily basis as the difference between the highest and lowest intraday prices scaled by the average. The sample period is from April 25, 2007, to June 26, 2007. Standard errors are clustered by day.

(b) Table 2: Stock-market reaction to the stamp-tax increase.

6.3 Table 3: Put warrant price and trading around May 30, 2007; Figure 3: Price and turnover of put warrants

Read:

- Table 3 shows very large increases in adjusted prices, turnover, trading volume, and volatility for put warrants after May 30.
- Figure 3 shows the timing and heterogeneity across individual put warrants. Turnover and prices jump immediately after the event.
- Because these put warrants were deep out of the money, the price increases are difficult to justify by fundamentals.

Economic message: The put-warrant evidence is the clearest bubble-amplification result: after the stock tax rose, speculative activity surged in an untaxed market with little fundamental value.

Table 3

Put warrant price and trading around May 30, 2007

A. Summary statistics								
	Mean	SD	P1	P25	P50	P75	P99	Obs.
Before May 30, 2007								
Price	1.16	0.48	0.37	0.98	1.22	1.27	1.99	98
BlackScholes_value	0.00	0.01	0.00	0.00	0.00	0.01	0.08	98
Adjusted_price	1.16	0.48	0.37	0.98	1.21	1.27	1.97	98
Turnover	67.10%	64.80%	15.70%	33.50%	45.80%	74.50%	385.40%	98
Volume (bn yuan)	0.29	0.13	0.09	0.20	0.25	0.39	0.64	98
log(Volume)	19.38	0.46	18.33	19.09	19.33	19.78	20.27	98
Volatility	5.50%	3.16%	1.64%	3.13%	4.54%	7.18%	16.00%	98
After May 30, 2007								
Price	3.57	1.94	0.11	1.96	3.49	5.13	8.15	96
BlackScholes_value	0.01	0.01	0.00	0.00	0.00	0.00	0.07	96
Adjusted_price	3.56	1.94	0.11	1.92	3.49	5.11	8.15	96
Turnover	559.20%	255.10%	204.90%	403.50%	515.50%	669.70%	1,741.00%	96
Volume (bn yuan)	11.31	10.34	1.21	5.57	8.09	12.22	45.68	96
log(Volume)	22.84	0.775	20.91	22.44	22.81	23.23	24.55	96
Volatility	41.80%	24.80%	11.00%	23.60%	33.30%	54.10%	150.50%	96
B. Regression results								
	(1)	(2)	(3)	(4)				
Dep. variable:	Adjusted_price	Turnover	log(Volume)	Volatility				
Post-event	2.405 (9.02)	4.344 (14.36)	3.303 (30.74)	0.328 (10.06)				
Maturity fixed effect	Yes	Yes	Yes	Yes				
Observations	194	194	194	194				
Adjusted R^2	.393	.701	.815	.583				

This table reports summary statistics (panel A) and results of regression analyses (panel B) of put warrants' price, turnover, volume, and price volatility before and after the event date of May 30, 2007. *Price* is a warrant's daily closing price. *BlackScholes_value* is a warrant's fundamental value based on the Black-Scholes model. *Adjusted_price* refers to a warrant's daily closing price minus its fundamental value based on the Black-Scholes model. *Volume* is daily trading volume in billion yuan. *log(Volume)* is the log of one plus daily trading volume (in yuan). *Turnover* equals the number of shares traded on each day divided by the number of outstanding shares. *Volatility* is measured on a daily basis as the difference between the highest and lowest intraday prices scaled by the average. In panel B, *Post-event* equals one for days on or after May 30, 2007 and otherwise zero. All regressions include maturity fixed effects. The sample period is from April 25, 2007, to June 26, 2007. Standard errors are clustered by day, and the corresponding *t*-statistics are reported in parentheses.

(a) Table 3: Put warrant price and trading.

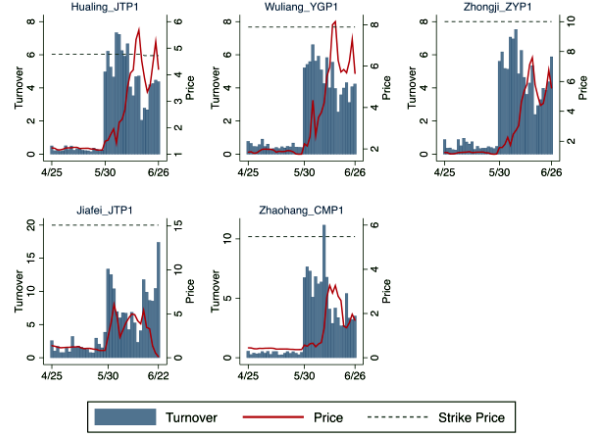


Figure 3

Price and turnover of put warrants around May 30, 2007

The figure plots daily turnover (bar, left y-axis), closing price (solid line, right y-axis), and strike price (adjusted for exercise ratio, dashed line, right y-axis) for each put warrant from April 25, 2007, to June 26, 2007.

(b) Figure 3: Put warrant turnover and price dynamics.

6.4 Table 4: Call warrant price and trading around May 30, 2007; Figure 4: Price and turnover of call warrants

Read:

- Table 4 shows that call warrants also experience higher turnover, volume, and volatility after the event.
- The call-warrant response is weaker than the put-warrant response because call warrants had more fundamental value and more heterogeneity.
- Interaction results show that more speculative call warrants before the event respond more strongly after the event.
- Figure 4 visually confirms the cross-warrant heterogeneity in speculative responses.

Economic message: The call-warrant evidence shows that the effect is not limited to a few nearly worthless put warrants. The migration channel also operates among call warrants, especially those with greater pre-event speculative activity.

Table 4
Call warrant price and trading around May 30, 2007

A. Summary statistics								
	Mean	SD	P1	P25	P50	P75	P99	Obs.
Before May 30, 2007								
Price	14.56	8.20	4.64	6.29	13.38	23.83	32.00	169
BlackScholes_value	14.64	9.54	3.87	6.64	9.77	25.90	36.13	169
Adjusted_price	-0.08	2.63	-5.34	-1.59	-0.31	0.77	6.54	169
Turnover	45.10%	26.90%	13.70%	25.70%	39.30%	56.90%	162.10%	169
Volume (bn yuan)	1.75	1.51	0.19	0.76	1.33	2.29	6.37	169
log(Volume)	20.98	0.80	19.04	20.45	21.01	21.55	22.57	169
Volatility	6.47%	3.06%	2.32%	4.27%	5.63%	8.11%	15.50%	169
After May 30, 2007								
Price	16.76	9.82	4.69	7.49	13.71	25.59	36.70	175
BlackScholes_value	16.23	12.09	3.09	7.30	10.81	27.20	40.04	175
Adjusted_price	0.53	3.74	-7.59	-1.61	0.74	2.45	9.77	175
Turnover	85.40%	50.90%	15.40%	48.00%	72.80%	113.30%	226.60%	175
Volume (bn yuan)	3.41	2.69	0.51	1.55	2.64	4.56	13.04	175
log(Volume)	21.68	0.751	20.05	21.16	21.7	22.24	23.29	175
Volatility	10.30%	4.54%	2.62%	6.89%	9.65%	13.40%	23.70%	175
B. Regression results								
	(1)	(2)	(3)	(4)				
Dep. variable:	Adjusted_price	Turnover	log(Volume)	Volatility				
Post-event	0.420 (1.65)	0.285 (4.50)	0.634 (7.21)	0.0306 (3.95)				
Maturity fixed effect	Yes	Yes	Yes	Yes				
Observations	344	344	344	344				
Adjusted R ²	.000	.086	.131	.130				

(a) Table 4 Panels A-B: Call warrant summary and baseline regressions.

Table 4
(Continued)

C. Interaction with warrant turnover				
	(1)	(2)	(3)	(4)
Dep. variable:	Adjusted_price	Turnover	log(Volume)	Volatility
Post-event	0.371	0.281	0.634	0.0306
	(1.49)	(4.39)	(7.18)	(3.91)
Post-event * Pre-event_turnover	7.523	1.325	0.876	0.0570
	(9.06)	(5.39)	(2.12)	(2.16)
Maturity fixed effect	Yes	Yes	Yes	Yes
Observations	344	344	344	344
Adjusted R ²	.360	.377	.142	.152
D. Interaction with warrant nominal price				
	(1)	(2)	(3)	(4)
Dep. variable:	Adjusted_price	Turnover	log(Volume)	Volatility
Post-event	0.445	0.289	0.637	0.0307
	(1.84)	(4.53)	(7.26)	(3.93)
Post-event * Pre-event_price	-0.117	-0.0253	-0.0208	-0.00098
	(-7.44)	(-5.93)	(-2.61)	(-2.18)
Maturity fixed effect	Yes	Yes	Yes	Yes
Observations	344	344	344	344
Adjusted R ²	.298	.328	.199	.139

This table reports summary statistics (panel A) and the results of regression analyses (panels B-D) of call warrants' price, turnover, volume, and price volatility before and after the event date of May 30, 2007. Price is ; warrant's daily closing price. BlackScholes_value is a warrant's fundamental value based on the Black-Schole model. Adjusted_price refers to a warrant's daily closing price minus its fundamental value based on the Black Scholes model. Volume is daily trading volume in billion yuan. log(Volume) is the log of one plus daily trading volume (in yuan). Turnover equals the number of shares traded on each day divided by the number of outstanding shares. Volatility is measured on a daily basis as the difference between the highest and lowest intraday price

(b) Table 4 Panels C-D: Speculativeness interaction and robustness.

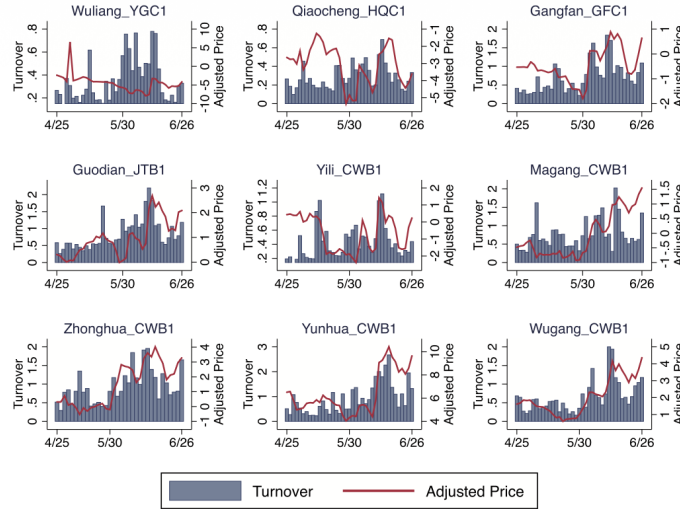


Figure 4
Price and turnover of call warrants around May 30, 2007
The figure plots daily turnover (left y-axis) and adjusted price (nominal price minus Black-Scholes value, right y-axis) for each call warrant from April 25, 2007, to June 26, 2007.

Figure 4: Call warrant turnover and price dynamics.

6.5 Table 5: Trading of stocks and warrants around May 30, 2007; Figure 5: First-time warrant investors

Read:

- Table 5 summarizes investor trading in stocks and warrants around the tax event.
- Figure 5 shows a sharp increase in the number of investors trading warrants for the first time after May 30.
- The entry of new warrant investors is consistent with substitution from the now-more-expensive stock market into the untaxed warrant market.

Economic message: These visuals identify the first mechanism: a stock-market tax shock redirected marginal investors into warrants.

Table 5
Investors' trading of stocks and warrants around May 30, 2007

	Mean	SD	P1	P25	P50	P75	P99	No. of investors
Before May 30, 2007								
<i>Number of total trades</i>	7.52	17.40	1.00	2.00	4.00	8.00	56.00	12,888,682
<i>Number of trades in stocks</i>	7.29	15.97	1.00	2.00	4.00	8.00	54.00	12,888,682
<i>Number of trades in warrants</i>	0.23	6.59	0.00	0.00	0.00	0.00	5.00	12,888,682
<i>Number of trades in puts</i>	0.08	3.03	0.00	0.00	0.00	0.00	2.00	12,888,682
<i>Number of trades in calls</i>	0.15	5.37	0.00	0.00	0.00	0.00	3.00	12,888,682
After May 30, 2007								
<i>Number of total trades</i>	7.59	21.32	0.00	1.00	3.00	8.00	70.00	12,888,682
<i>Number of trades in stocks</i>	6.11	12.97	0.00	1.00	3.00	7.00	50.00	12,888,682
<i>Number of trades in warrants</i>	1.49	16.22	0.00	0.00	0.00	0.00	37.00	12,888,682
<i>Number of trades in puts</i>	1.21	13.12	0.00	0.00	0.00	0.00	31.00	12,888,682
<i>Number of trades in calls</i>	0.28	6.51	0.00	0.00	0.00	0.00	7.00	12,888,682

This table reports summary statistics of investors' number of trades in stocks and warrants before and after the event date of May 30, 2007. Over the 20 trading days before or after the event, each investor's number of total trades, trades of stocks, warrants, puts, and calls are counted.

(a) Table 5: Investor trading summary.

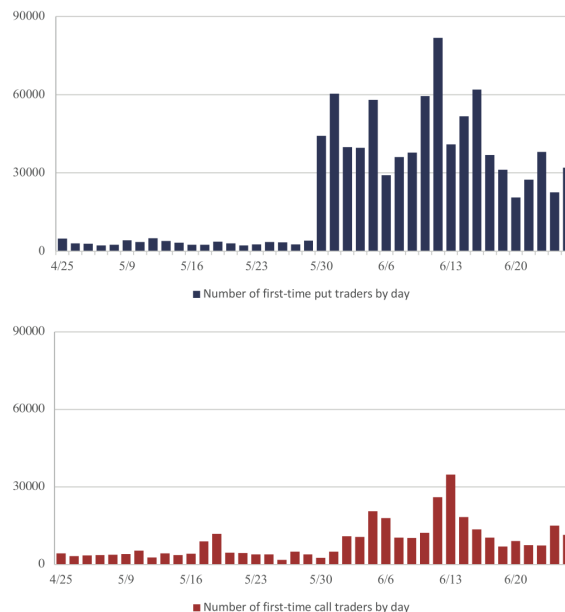


Figure 5
Number of first-time warrant traders by day
The upper (lower) panel plots the daily number of first-time put (call) trader from April 25, 2007, to June 26, 2007. For each day, first-time traders are defined as stock investors who start trading call or put warrants for the first time on that day.

(b) Figure 5: First-time warrant investors.

6.6 Figure 6: Investors by activeness group; Figure 7: Stock and warrant trading around May 30

Read:

- Figure 6 sorts warrant investors by their pre-event activeness and shows that active investors trade far more intensively after the event.
- Figure 7 compares trading behavior in stocks and warrants around the event for different investor groups.
- Active traders' warrant purchases and sales rise strongly after the tax shock.

Economic message: These figures identify the second mechanism: the tax shock not only brought in new investors, but also intensified speculative trading by existing warrant traders.

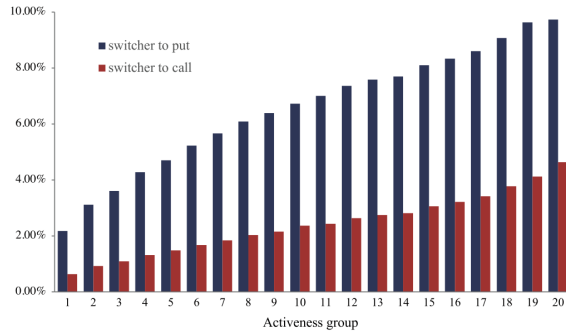
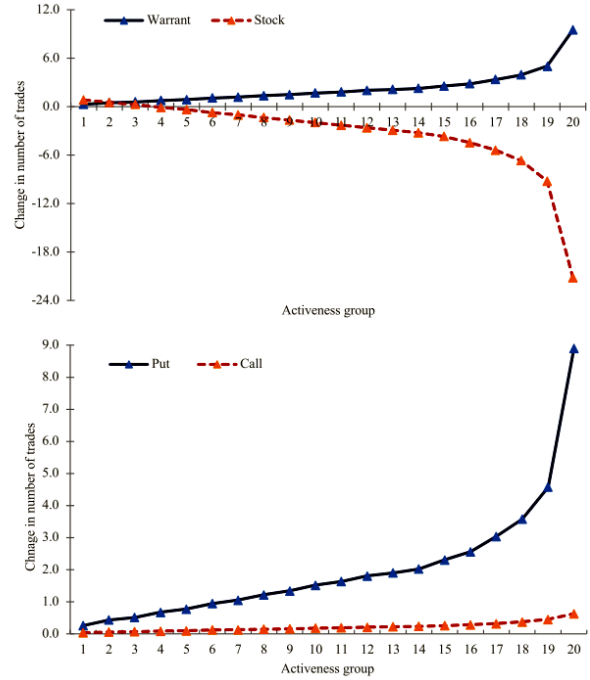


Figure 6
Switchers from stock trading to warrant trading
The event window is from April 25, 2007, to June 26, 2007. *Switcher* is defined as a stock investor who does not trade any warrants during the event window before May 30 but starts to trade warrants after. The figure plots the fraction of the number of *Switchers* after May 30, 2007, to the total number of investors in each activeness group. The 20 activeness groups are sorted by investors' total number of trades before the event.

(a) Figure 6: Warrant investors by activeness group.



(b) Figure 7: Stock and warrant trading by investors around May 30.

6.7 Table 6: Regression results for warrant investor trading; Table 7: Overnight returns and new warrant investors

Table 6
Substitution of trading and investor activeness

	(1)	(2)	(3)	(4)
Dep. variable: Number of trades	Warrant	Warrant	Stock	Stock
<i>Post-event</i>	1.2547 (257.23) [3.9]	-0.2067 (-28.59) [-0.92]	-1.1784 (-205.63) [-1.66]	2.0254 (278.22) [3.54]
<i>Activeness</i>		0.0827 (129.15) [2.21]		1.6634 (2581.15) [4.25]
<i>Activeness*Post-event</i>		0.2467 (272.62) [4.88]		-0.5409 (-593.56) [-4.16]
Observations	25,777,364	25,777,364	25,777,364	25,777,364
Adjusted R^2	.0026	.0133	.0016	.2745
	(5)	(6)	(7)	(8)
Dep. variable: Number of trades	Call	Call	Put	Put
<i>Post-event</i>	0.1263 (53.70) [4.67]	0.0052 (1.47) [0.49]	1.1284 (300.77) [3.82]	-0.2118 (-38.08) [-0.98]
<i>Activeness</i>		0.0554 (178.83) [2.16]		0.0273 (55.43) [2.33]
<i>Activeness*Post-event</i>		0.0205 (46.70) [8.70]		0.2263 (324.98) [4.69]
Observations	25,777,364	25,777,364	25,777,364	25,777,364
Adjusted R^2	.0001	.0037	.0035	.0011

This table presents regressions of investors' number of trades in stocks, warrants, calls, and puts on *Post-event* and investor activeness. The dependent variable is the number of trades in warrants (columns 1 and 2), stocks (columns 3 and 4), calls (columns 5 and 6), and puts (columns 7 and 8), over the 20 trading days before or after the May 30, 2007, event. *Post-event* is a dummy variable that equals one if it is after the event. *Activeness* refers to the semidecile score of investors' activeness groups, based on investors' total number of pre-event trades. The *t*-statistics reported in parentheses are based on standard errors clustered by investor, and *t*-statistics in brackets are based on standard errors clustered by activeness group.

Table 7
Overnight and daytime returns of stocks and warrants on May 30, 2007

Warrant code	Warrant name	Warrant			Underlying stock		Implied delta	Extrapolated daytime returns	Realized minus extrapolated
		Overnight	Daytime	Total	Overnight	Daytime			
Call warrants									
30002	Wailing YGC1	-5.7%	-1.3%	-6.9%	-6.4%	2.0%	0.89	1.8%	-3.1%
31001	Qiaocheng HQC1	-9.0%	-2.0%	-10.8%	-8.1%	-1.0%	1.11	-1.1%	-1.0%
31002	Guangfu GDC1	-	-	-	-	-	-	-	-
38008	Guodian JTB1	-	-	-	-	-	-	-	-
38009	Yib CWB1	-7.8%	-1.6%	-9.3%	-9.9%	3.2%	0.79	2.5%	-
38010	Majang CWB1	-8.4%	-4.2%	-12.3%	-4.9%	-3.4%	1.71	-9.2%	5.0%
38011	Zhanghua CWB1	-7.9%	-1.8%	-15.1%	-8.3%	-1.8%	0.94	-1.7%	-6.1%
38012	Yushui CWB1	-7.4%	-4.4%	-11.5%	-6.0%	-4.0%	1.24	-5.0%	0.5%
38013	Wugang CWB1	-6.4%	-5.9%	-11.9%	-5.8%	-4.5%	1.10	-4.9%	-1.0%
	Average	-7.5%	-3.9%	-11.1%	-7.1%	-1.6%	1.11	-2.5%	-1.4%
Put warrants									
38097	Zhanghang CMP1	4.8%	46.7%	53.8%	-5.1%	-1.1%	-0.94	1.1%	45.6%
38003	Huailin JTP1	4.8%	22.1%	28.0%	-7.4%	-2.8%	-0.65	1.8%	20.3%
38004	Wailing YGP1	8.5%	18.3%	28.3%	-6.4%	2.0%	-1.32	-2.7%	21.0%
38006	Zhongji ZTP1	6.5%	31.6%	37.5%	-9.9%	2.7%	-0.46	-1.2%	32.8%
38008	Jiushi JTP1	6.9%	171.6%	190.3%	-2.4%	-6.4%	-2.47	18.4%	153.2%
	Average	5.9%	58.1%	67.6%	-6.3%	-1.1%	-1.23	3.5%	54.6%
All stocks									
		Overnight	Daytime	Total					
	Average	-7.3%	-1.0%	-8.2%					
	SD	2.6%	3.2%	3.6%					

This table presents the returns of warrants, warrants' underlying stocks, and all stocks on May 30, 2007. Overnight return is defined as the percentage change of the open price on May 30 to the closing price on May 29. Daytime return equals the percentage change of the closing price on May 30 to the open price. Total return is the sum of overnight and daytime returns. Implied delta is calculated as the ratio of the warrant overnight return to the corresponding underlying stock return. Extrapolated daytime return equals implied delta multiplied by the daytime return of underlying stocks. Realized minus extrapolated refers to the difference between the warrant daytime return and the extrapolated daytime return. Call warrants 31002 and 38006 were suspended for trading on May 30, 2007.

Read:

- Table 6 confirms the visual evidence with regressions: warrant trading intensity rises after the event, especially among active investors.

- Table 7 links overnight returns to the arrival of new investors and implied delta calculations.
- The evidence is strongest for put warrants, where the fundamental value channel is weakest.

Economic message: The tables connect investor flows to price pressure. They suggest that new investor demand at the open helped push warrant prices up, while active traders amplified intraday turnover.

6.8 Table 8: Placebo events; Table 9: Investor trading on and prior to May 30, 2007

Read:

- Table 8 compares May 30 with other large stock-market decline days. The placebo events do not generate the same strong pattern of warrant entry and trading.
- Table 9 shows that active investor groups greatly increased trading intensity on May 30 relative to the pre-event period.
- The most active investors have extremely high trading frequencies, suggesting intense speculative rotation rather than passive hedging.

Economic message: The placebo and investor-group evidence strengthens the causal interpretation: May 30 is not just a large down day; it is a regulatory shock that redirected and amplified speculation.

Table 8
Return of stocks and warrants on placebo event days

	Market return	Return of calls	Return of puts	Number of first-time put traders
<u>May 30, 2007</u>				
Mean	-7.02%	-11.1%	67.6%	44,239
SD		2.53%	69.4%	
Observations		7	5	
<u>Placebo dates</u>				
February 27, 2007	-8.96%	-16.7%	5.27%	3,223
June 7, 2006	-6.05%	-10.1%	1.31%	6,581
July 5, 2007	-5.50%	-6.60%	-10.6%	10,089
July 13, 2006	-5.34%	-0.23%	23.3%	5,180
January 31, 2007	-5.19%	-7.68%	-7.17%	1,059
<u>Placebo date summary</u>				
Mean	-6.21%	-8.63%	5.31%	5,226.4
SD		7.04%	19.2%	
Observations		57	62	
Difference = Event - Placebo		-2.48%	62.3%	
t-statistics		(-0.92)	(5.28)	

This table summarizes the returns for the stock market, call warrants, and put warrants, and the number of first-time put traders on May 30, 2007, and on the five placebo event dates. The five placebo event dates are the days with the lowest market returns between December 5, 2005, to October 31, 2007, excluding April 25 to June 26, 2007. Market return is calculated with the value-weighted return of all A-share stocks.

Table 9
Investor trading on and prior to May 30, 2007

A. Put warrants										
	First time		Top 0.1%		0.1%-1%		1%-5%		5%-50%	
(per day)	On event	Pre-event	On event	Pre-event	On event	Pre-event	On event	Pre-event	On event	Pre-event
Number of trades	4.0	14.2	118.3	2.0	36.8	0.6	17.8	0.2	5.6	0.1
Number of purchases	2.2	6.4	52.8	1.0	18.4	0.3	9.1	0.1	2.9	0.1
Number of sales	1.7	7.7	65.5	1.0	18.4	0.3	8.7	0.1	2.7	0.0
Total value of trades	172.6	1750.7	34572.6	135.0	3887.4	29.0	1065.8	7.4	252.1	3.4
Total value of purchases	95.5	876.2	17211.8	68.2	1929.9	14.6	510.7	3.9	125.6	1.8
Total value of sales	79.1	876.5	17360.8	66.8	1958.4	14.4	554.1	3.5	126.5	1.6
# of investors	44,239	131	1,185	1,185	5,265			59,232		65,814
B. Call warrants										
	First time		Top 0.1%		0.1%-1%		1%-5%		5%-50%	
(per day)	On event	Pre-event	On event	Pre-event	On event	Pre-event	On event	Pre-event	On event	Pre-event
Number of trades	2.0	105.7	179.4	14.7	34.7	3.9	11.0	0.9	3.1	0.4
Number of purchases	1.4	48.1	83.6	7.0	18.0	2.0	6.1	0.5	1.7	0.2
Number of sales	0.6	57.5	95.9	7.7	16.7	1.9	4.9	0.4	1.5	0.2
Total value of trades	181.0	20805.4	55164.7	8636.7	9877.2	877.4	1944.6	93.0	548.9	27.7
Total value of purchases	49.0	13402.5	27475.9	1818.6	4623.9	445.8	1022.0	47.7	167.7	20.0
Total value of sales	2.99	27	27	1820.1	4433.2	431.6	942.6	45.3	181.2	17.7
# of investors	2,599	27	247	247	1,100			12,580		15,744

Investors trading put or call warrants are sorted based on the Number of trades in puts or calls on May 30 into five groups: the top 0.1%, 0.1% to 1%, 1% to 5%, 5% to 50%, and the bottom 50%. Number of trades equals the daily number of trades in puts or calls. Number of purchases/sales equals the number of purchases/sales in puts or calls per day. Total value of trades/purchases/sales is the total value of investors' daily trades/purchases/sales (in thousands) in puts or calls. Columns labeled with "Pre-event" report the mean of the statistics over the 20-day window prior to May 30, 2007. Columns labeled with "On event" report the mean of statistics on May 30, 2007. Column "First time" refers to the group of investors who traded put or call warrants for the first time on May 30. Panel A reports the statistics for put warrants, and panel B reports the statistics for call warrants. The sample includes all investors who trade warrants on the event day.

7 Short summary

- The Chinese government tripled the stamp tax on stock trading on May 30, 2007.
- The tax modestly reduced activity in the stock market, the intended target.
- Warrants were not taxed, and investors could use them as speculative substitutes.
- Put warrants, which were deep out of the money and had little fundamental value, experienced sharp increases in adjusted prices, turnover, volume, and volatility.
- Call warrants also experienced increases, especially those that were more speculative before the event.
- New investors entered the warrant market after the tax shock.
- Existing active warrant investors also increased trading intensity substantially.
- Placebo large stock-market drops do not show the same pattern, supporting the interpretation that the tax shock was special.
- The evidence illustrates a regulatory whack-a-mole problem: suppressing speculation in one market may push it into a nearby untaxed market.

8 Conclusion

8.1 Core conclusion

The paper shows that a Tobin tax can have unintended cross-market effects. China's May 30, 2007, stock stamp-tax increase reduced stock-market activity but intensified speculative trading in the untaxed warrant market. The policy partially solved one problem while worsening another.

8.2 Economic intuition

Transaction taxes change relative trading costs. When speculative investors have access to substitute instruments, they can respond by moving to the lower-cost venue. If the substitute market is less regulated, less fundamentally anchored, or more prone to bubbles, the policy can increase rather than decrease speculative excess.

8.3 Policy implication

The central regulatory lesson is that financial policies should be designed with substitution margins in mind. A tax or restriction aimed at one asset class can fail if investors can replicate speculative exposure elsewhere. Effective regulation therefore requires monitoring the full ecosystem of related instruments, not only the market directly targeted by the rule.

8.4 Limitations and external validity

The warrant market in China was unusually speculative, short-sale constrained, and retail dominated. Therefore, the exact magnitude of the effect should not be generalized mechanically to all transaction taxes. The broader mechanism, however, is portable: regulation can redirect activity to untaxed markets when substitution is easy.

8.5 Takeaway

The paper's most important message is not that Tobin taxes never work. It is that Tobin taxes can trigger investor migration and spillovers. Regulators who ignore these spillovers risk playing a whack-a-mole game: each intervention may push speculative activity into a new corner of the financial system.